The Flag Container

why it gives Demazure operators and not Schubert polynomials

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Abstract

The Targets table of `q-container.pdf` has rows for the q-Leibniz rule, q-Pascal, Pieri–Littlewood–Richardson and Macdonald polynomials, and no row for Schubert polynomials. This note attempts to earn one and instead establishes that the row cannot be written in the form expected.

Schubert polynomials are generated from the top class by divided differences $\partial_i f = (f - s_i f)/(x_i - x_{i+1})$, and the container derivative of `q-container.pdf` Theorem D is the Jackson operator $D_q f = (f - \sigma_q f)/((1-q)x)$. Both have the shape (1 *minus a group element*) over (*a difference of weights*), Jackson dividing by a scaling and Demazure by a transposition; and a container already carries transpositions, since relabelling positions is what a container's symmetry *is*. The conjecture was therefore that ∂_i needs no new machinery.

Half of that survives. Building the operator from the fibration that forgets the i-th subspace of a flag, we verify by exact enumeration ($n \le 4$, $q = 2, 3$) that it descends to the cell module and satisfies the *braid* relations, so $\mathfrak{S}_w$ is independent of the reduced word at container level with no new machinery, as predicted. But the operator satisfies the *Hecke* quadratic and not the nilCoxeter relation $\partial_i^2 = 0$; and we prove (Theorem 4.2) that no operator attached to a single index can be nilpotent unless $q = -1$. The container hands you an idempotent where Schubert polynomials need a nilpotent, and an idempotent divided difference is a *Demazure* operator, whose polynomials are key polynomials.

The consequence is a correction to the dictionary rather than an addition to it: counting $\mathbb{F}_q$-points sees the Hecke algebra and the Poincaré polynomial $\sum_w q^{\ell(w)} = [n]_q!$, while divided differences live in the Chow ring, which is a different decategorification of the same cells.

Companion: `q-container.pdf`; its remark that the Hecke algebra is the endomorphism algebra of the flag container is the starting point. Code: `code/flag.py`, results recorded in `code/FINDINGS.md`.

1 The question, and why it is not idle

`q-container.pdf` observes that $\mathrm{Ind}_B^G \mathbf{1} = k[G/B]$ is a free module on a hom-set — a container in the $k[-]$ sense — and that $\mathrm{End}_G(\mathrm{Ind}_B^G \mathbf{1}) = H_n(q)$, so that *the Hecke algebra is the endomorphism algebra of the flag container*. Since the Schubert basis of $H^*(\mathrm{Fl}_n)$ is indexed by the same Weyl group that indexes $H_n(q)$, it is natural to expect the Schubert calculus to be visible from the container side.

The expectation has a precise form. Write ∂_i for the divided difference and D_q for the Jackson q-derivative of Theorem D:

$$\partial_i f = \frac{f - s_i f}{x_i - x_{i+1}}, \qquad D_q f = \frac{f - \sigma_q f}{(1-q)\,x}, \quad \sigma_q f(x) = f(qx).$$

Both are *divided differences for a group element*: a difference operator over the corresponding difference of weights. Theorem D says the container derivative is the Jackson one, where the

group element is the scaling σ_q. Containers carry the symmetric group on positions for free, so nothing new should be needed to build the operator with s_i in place of σ_q.

Conjecture 1.1 (under test)**.** *The flag container carries operators ∂_i, built from its own structure, generating the nilCoxeter algebra: $\partial_i^2 = 0$, together with the braid relations, so that $\mathfrak{S}_w = \partial_{w^{-1}w_0}(x^\delta)$ is well defined.*

Remark 1.2 (what would make this circular). Defining ∂_i by its action on Schubert classes — $\partial_i \mathfrak{S}_w = \mathfrak{S}_{ws_i}$ when $\ell(ws_i) < \ell(w)$ and 0 otherwise — and then verifying the nilCoxeter relations would verify nothing. Below the operator is built from the geometry and its relations are read off.

2 The flag container

Definition 2.1 (flag container)**.** Let $\mathrm{Fl}_n(\mathbb{F}_q)$ be the set of complete flags $0 = F_0 < F_1 < \cdots < F_n = \mathbb{F}_q^n$, and E the standard flag. The *relative position* of F to E is the permutation $w \in S_n$ determined by the rank matrix $r_{ij} = \dim(E_i \cap F_j)$ via

$$w(j) = i \iff r_{ij} - r_{i-1,j} - r_{i,j-1} + r_{i-1,j-1} = 1.$$

The *flag container* is $S_n \triangleleft \left(w \mapsto \mathbb{F}_q^{\ell(w)}\right)$, with extension $[\![\cdot]\!]X = \sum_{w \in S_n} X^{\ell(w)}$.

So the shapes are the permutations and the positions over shape w are the cell C_w. This is a q-container in the sense of `q-container.pdf` Definition 2: the shapes form a set, the positions are $\mathbb{F}_q$-spaces.

Proposition 2.2 (Bruhat, by enumeration)**.** *$|\mathrm{Fl}_n(\mathbb{F}_q)| = [n]_q!$ and $|C_w| = q^{\ell(w)}$ for every w.* ***confirmed*** *for $(n, q) \in \{(2,2), (2,3), (3,2), (3,3), (4,2)\}$, computed from the rank matrix of Definition 2.1 rather than assumed.*

n	q	flags	$[n]_q!$	cells
2	2	3	3	2
2	3	4	4	2
3	2	21	21	6
3	3	52	52	6
4	2	315	315	24

3 The operator, built from the geometry

Definition 3.1 (the fibration, and its operator)**.** Let π_i forget the i-th subspace of a flag. Define S_i on functions on $\mathrm{Fl}_n(\mathbb{F}_q)$ by pushforward followed by pullback:

$$(S_i f)(F) \;=\; \sum_{\pi_i(F') = \pi_i(F)} f(F').$$

Nothing about Schubert cells enters. Two things then have to be checked rather than assumed: that the fibre is what it should be, and that S_i descends to the cell module at all.

Proposition 3.2. *Every fibre of π_i has exactly $q + 1 = |\mathbb{P}^1(\mathbb{F}_q)|$ elements, and S_i applied to a cell indicator is constant on cells, hence descends to the free module on S_n.* ***confirmed****, 5/5 cases.*

Example 3.3 ($n = 3$, $q = 2$, the operator read off)**.** In the basis of cell classes,

$$S_1[w] \;=\; \begin{cases} [w] + [ws_1], & \ell(ws_1) > \ell(w), \\ q[w] + q[ws_1], & \ell(ws_1) < \ell(w). \end{cases}$$

For instance $S_1[123] = [123] + [213]$ and $S_1[213] = 2[123] + 2[213]$.

4 What it satisfies, and what it does not

Proposition 4.1. *With $T_i := S_i - I$:*
(a) $S_i^2 = (q+1)S_i$, so $S_i/(q+1)$ is idempotent;
(b) $T_i^2 = (q-1)T_i + qI$, the Hecke quadratic;
(c) $T_iT_jT_i = T_jT_iT_j$ for $|i-j| = 1$, and $T_iT_j = T_jT_i$ for $|i-j| \geq 2$;
(d) $T_i^2 \neq 0$ in every case tested.
confirmed *for (a)–(c), 5/5 cases, all 24 cells at $n = 4$.*

Part (c) is the half of Conjecture 1.1 that survives, and it survives exactly as predicted: the braid relations come from relabelling positions, need no new machinery, and are what makes $\mathfrak{S}_w$ independent of the reduced word.

Part (d) is the half that fails. It is worth being precise about how badly.

Theorem 4.2 (no divided difference at finite q)**.** *Fix i and let A_i be the subalgebra generated by T_i. Then $A_i = \mathrm{span}\{I, T_i\}$, and the only $u \in A_i$ with $u^2 = 0$ is $u = 0$, unless $q = -1$.*

Proof. $A_i = \mathrm{span}\{I, T_i\}$ because Proposition 4.1(b) expresses T_i^2 in that span, so the span is closed under multiplication; `flag.py` confirms T_i^2 and T_i^3 lie in it for all cases tested. Write $u = aT_i + bI$. Then

$$u^2 = a^2T_i^2 + 2ab\,T_i + b^2I = \big[a^2(q-1) + 2ab\big]T_i + \big[a^2q + b^2\big]I.$$

T_i and I are linearly independent, so $u^2 = 0$ forces both brackets to vanish. If $a = 0$ the second gives $b = 0$. If $a \neq 0$ the first gives $b = a(1-q)/2$, and substituting into the second,

$$a^2q + \tfrac{1}{4}a^2(1-q)^2 = 0 \iff 4q + (1-q)^2 = 0 \iff (q+1)^2 = 0. \qquad \square$$

Corollary 4.3. *No nilpotent divided difference exists on the flag container over any finite field, since $q \geq 2$.*

Remark 4.4 (the scope of the theorem, stated plainly). The theorem is about operators *attached to a single index*, which is what a ∂_i is. It does not claim the full algebra has no nilpotent elements — $H_n(q)$ is a matrix algebra for generic q and has many. What it claims is that none of them is a ∂_i, and the two-dimensionality of A_i is what makes the affine search exhaustive rather than a sample. A box search over $|a|, |b| \leq 6$ independently returns nothing in all five cases.

Refuted conjecture 4.5. Conjecture 1.1 is **refuted** in its nilCoxeter half. The flag container produces $H_n(q)$, whose $q \to 0$ degeneration is the 0-Hecke algebra of *idempotent* operators, not the nilCoxeter algebra of nilpotent ones.

operator	relation	generates
$S_i/(q+1)$	idempotent	Demazure operators, **key polynomials**
$T_i = S_i - I$	Hecke quadratic	$H_n(q)$, Hall–Littlewood, Green
∂_i	nilpotent	**Schubert polynomials** — not reachable here

5 What the Schubert basis is, at container level

The refutation is not a dead end; it relocates the object.

Proposition 5.1. *The flag container is $\sum_{w \in S_n} X^{\ell(w)}$, so a Schubert class is a shape of the flag container together with its position space $\mathbb{F}_q^{\ell(w)}$, and the one-variable shadow of $\mathfrak{S}_w$ is $x^{\ell(w)}$. Decategorifying the whole container gives the Poincaré polynomial $\sum_w x^{\ell(w)}$, which at $x = q$ is $[n]_q!$.*

This says why the divided differences were never going to appear. The multivariable $\mathfrak{S}_w(x_1, \ldots, x_n)$ records the *torus weights* on positions; a point count records only how many positions there are. The two decategorifications of the same cell decomposition are

$$\text{count } \mathbb{F}_q\text{-points} \longrightarrow H_n(q),\ [n]_q! \qquad \text{versus} \qquad \text{Chow ring} \longrightarrow \partial_i,\ \mathfrak{S}_w,$$

and `q-container.pdf` is built on the first.

Question 5.2 (the row the Targets table is missing). Grade the positions of the flag container by the T-action, so that a position carries a weight rather than only contributing to a dimension. Does the multigraded flag container have $\mathfrak{S}_w(x_1, \ldots, x_n)$ as its shape polynomial, and do the divided differences appear as operations on the grading?

Remark 5.3 ($q = -1$). Theorem 4.2 fails exactly at $q = -1$, which is not a prime power and so lies outside the theory rather than inside it at a special point. We record the value and claim nothing from it.

6 Status of every claim

#	claim	prior	verdict
5	Bruhat decomposition, by enumeration	—	**confirmed**, 5/5
6	braid relations hold at container level	70%	**confirmed**, $n \leq 4$
7	$\partial_i^2 = 0$ at container level	60%	**refuted**, Thm 4.2
8	multigraded flag container gives $\mathfrak{S}_w(x)$	new	open, Q 5.2

Independently of the container side, all 6 and all 24 Schubert polynomials for S_3 and S_4 were computed in $\mathbb{Z}[x]$ by divided differences from x^δ, with agreement across reduced words asserted at every step; this is the control, and it confirms $\mathfrak{S}_{132} = x_1 + x_2$ and $\mathfrak{S}_{321} = x_1^2 x_2$ against the standard tables.

Reproducing. `python3 code/flag.py`. Exact enumeration over $\mathbb{F}_2$ and $\mathbb{F}_3$ for $n \leq 4$; no sampling anywhere.